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Spatial Meets AI
Intelligent Geospatial Workflows with Oracle

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Image by Gemini

Claude
Opus 5.5 Extra
Anthropic

Our Journey Today

1. Where It Starts
2. Oracle Spatial *(not „Special“)*
3. Spatial Meets AI: Two Directions
4. Takeaways

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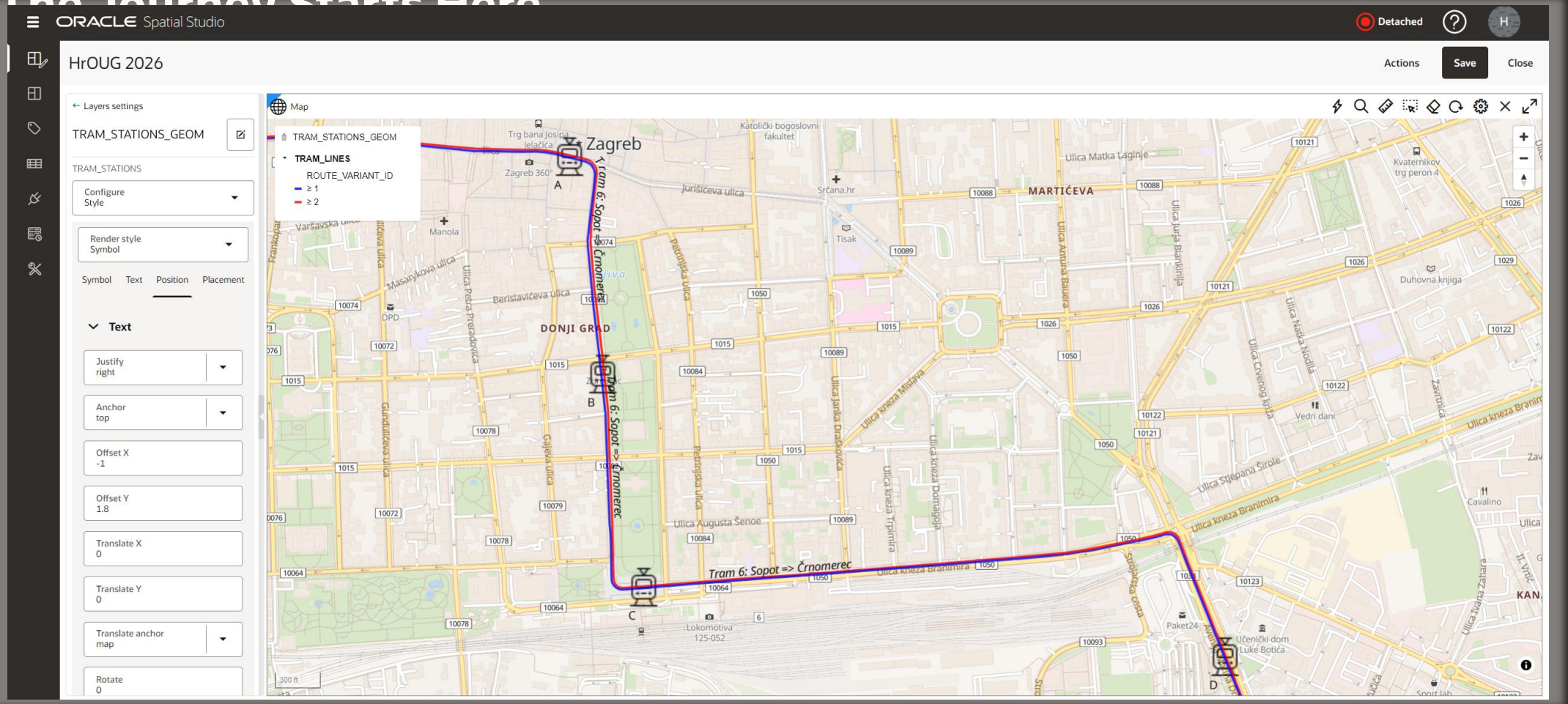
Don´t Panick

Scripts used in the session will be published!
They were generated using Claude.



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The Journey Starts Here



Oracle Spatial Studio project

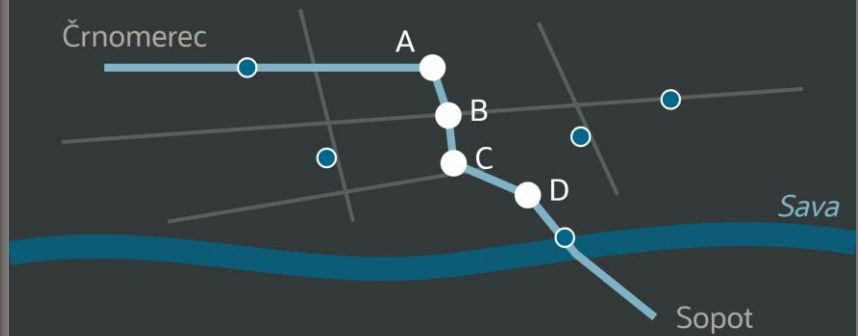
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Hot Stops, Cool Stops

It's been a while, but I'm sure you remember

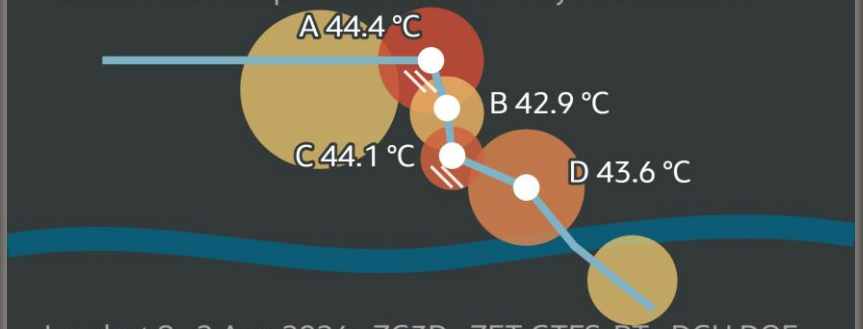
3 pm · July · Tram 6

Where would you wait for tram 6?



Surface temperature + shade at 3 pm

Land surface temperature at 11:39 · city mean 38.2 °C



Landsat 8 · 2 Aug 2026 · ZG3D · ZET GTFS-RT · DGU DOF

You know four stops by heart. Zagreb has

more than 1000

Which first? For whom? How much shade?

25 minutes · one database

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Oracle Spatial

The Foundation – In every edition of the Oracle Database wherever it is deployed

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The Spatial Foundation in the Oracle AI Database



Vector

Points, lines, areas



Point clouds

Laser scanning, photogrammetry



Raster

Imagery, terrain models, statistical grids



Network Data Model

Routing, tracing



Addresses

Geocoding and reverse geocoding



Linked Data

Semantic web support



3D

Solids, buildings, city models



Interactive visualization

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Recent Spatial Capabilities

Data loading

SDO_GEOMETRY(lon, lat): one constructor for WGS84 points (2001 / 4326)

Predefined constants: SDO_POINT2D, SDO_POLYGON2D, SDO_LONLAT, SDO_WEBMERCATOR instead of magic numbers

Automatic spatial metadata: created together with the spatial index, with no manual USER_SDO_GEOM_METADATA insert

Oracle Spatial · Vector data

Spatio-temporal queries on point data: the CBTREE index includes date and timestamp columns (FILTER BY), so one index search covers place and time (RU 23.26.3)

H3 indexes: hexagonal aggregation of large point volumes

Oracle Spatial · 3D

PC_DIFFERENCE: detect changes between two point clouds

GENERATE_CROSS_SECTION_AS_GEOMS: cross-sections through a point cloud

CREATE_MESHES: 3D meshes from TINs

ORDS · Serving

Vector tiles in SQL: SDO_UTIL.GET_VECTORTILE, Mapbox-compatible, straight to MapLibre or APEX

Vector tile cache: pre-rendered tiles for faster maps

H3 tiles: SDO_UTIL.H3SUM_VECTORTILE for aggregated maps

Spatial Studio

Built into Autonomous AI Database: no separate deployment

AI Labeling: create training labels for segmentation and object detection on imagery, and export them in standard formats

Coming: train and use these imagery models in Spatial AI

Select AI · MCP Server · AI Vector Search

Select AI Agent: tools that wrap SQL, PL/SQL, NL2SQL and RAG, so spatial logic becomes an agent tool

Autonomous AI Database MCP Server: built in and managed; any MCP client can call those tools, with database security and auditing

Multimodal embeddings: CLIP image and text models loaded through the OML4Py ONNX pipeline, with large ONNX models supported *(23.26.1)*

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Demo Part 1: Stream ZET Vehicle Positions on Maps

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Spatial Meets AI

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Spatial Meets AI: Two Directions

One governed foundation: Oracle AI Database 26ai

AI for Spatial Data

Turn imagery and location data into features, patterns and predictions.

- **Detect, classify, extract:** label imagery for segmentation and object detection, and turn the results into geometries. *(Spatial Studio in Autonomous AI Database, AI Labeling, GeoRaster)*
- **Find patterns that only exist in space:** hotspots, colocation, local drivers, regions. *(Spatial AI in OML4Py)*
- **Run models where the data lives:** save once, call from SQL and REST. *(OML, Embedded Python Execution)*

Our Zagreb example: *Where is it hot, and why?*

Spatial in AI Agents

Give AI agents location as context, and spatial logic as tools.

- **Ask in plain language:** questions become SQL over curated spatial views. *(Select AI)*
- **Act through tools:** spatial logic wrapped as tools an agent can call. *(Select AI Agent)*
- **Plug into any agent:** the same tools, available to any MCP client, under database security and auditing. *(Autonomous AI Database MCP Server)*

Our Zagreb example: *“I have budget for 20 shelters and 50 trees. Where should they go?”*

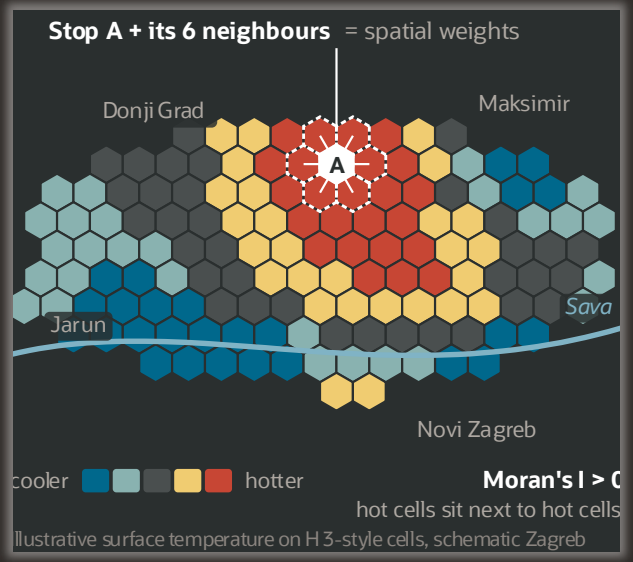
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First Direction: AI for Spatial Data

Details: docs.oracle.com/en/cloud/paas/autonomous-database/serverless/cspai/oracle-spatial-ai1.html

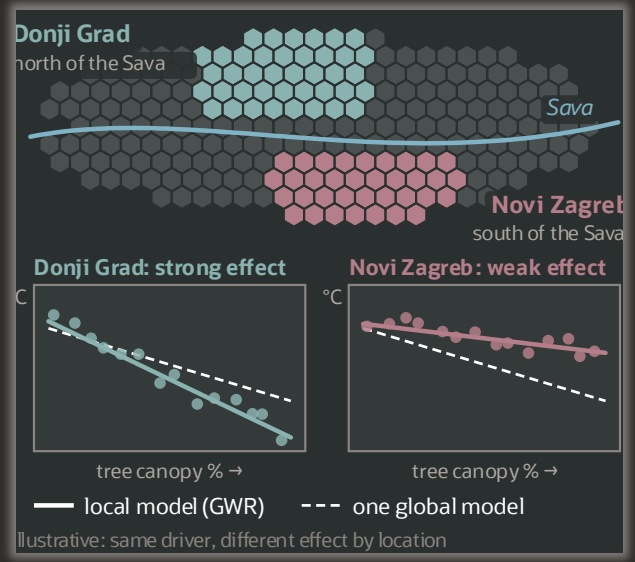
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Spatial ML in 5 Minutes



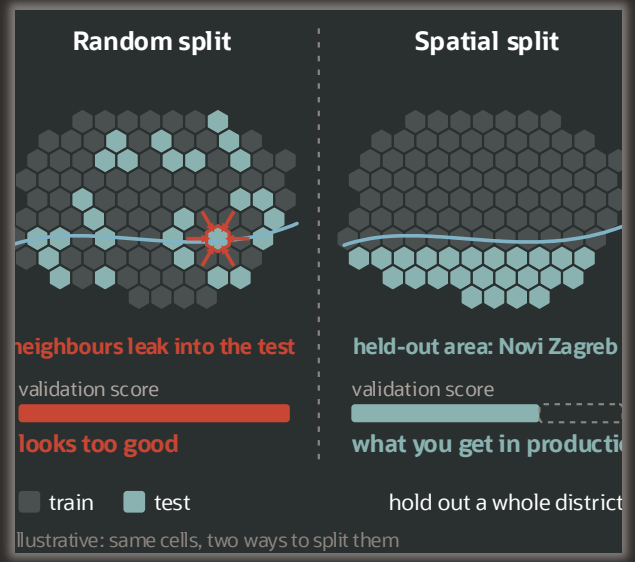
Spatial Dependence

Near things are more alike. A hot stop usually has hot neighbours.
→ Measure it with spatial weights (k-nearest neighbours, distance band, kernel) and Moran's I.



Spatial Heterogeneity

The cooling effect of trees isn't the same everywhere in Zagreb. A tree on Trg bana Jelačića is worth more than a tree in Maksimir.
→ Use local models (GWR, geographical models, spatial regimes) instead of one global model.



Spatial Leakage

A random train/test split puts neighbours on both sides, so accuracy looks better than it is.
→ Use spatial train/test splits and spatial pipelines.

Images generated using Claude

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Spatial ML in 5 Minutes

How Oracle Spatial AI handles it

Question	Technique	Oracle Spatial AI (OML4Py)
Is it clustered at all?	Global spatial autocorrelation	Moran's I
Where are the hot and cold spots?	Local Moran's I (LISA)	`LISAHotspotClustering`
Do two things occur together?	Local colocation quotient	`spatial_colocation_analysis`
What drives it, and does that vary by place?	OLS diagnostics → lag/error model → local regression	`OLSRegressor`, Spatial Lag / Error Model, `GWRRegressor`, Spatial Regimes, Fixed Effects
Which areas belong together?	Regionalization	`AgglomerativeClustering`, DBSCAN with regionalization, K-Means
What's unusual?	Spatial outlier detection	`LocalOutlierFactor`
Which class does it belong to?	Spatial classification	SLX, Geographical and GWR classifiers



Run the Machine Learning where the data resides.



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Spatial AI Algorithms and What They Tell You

Question	Spatial AI component	Insight for Zagreb
Is heat spatially structured at all?	Global Moran's I with KNN or contiguity weights	Strong autocorrelation, so plain regression is biased.
Where are the significant heat and cool islands?	`LISAHotspotClustering`	High-high heat islands; low-high cool islands (large parks, Jarun, the Sava banks); outliers such as a cool park inside a hot core.
Do vulnerable people live near the hottest stops?	`spatial_colocation_analysis` (local colocation quotient, optional time window)	Stops significantly collocated with residents aged 65+ or with care facilities.
What drives surface temperature, and does heat spill over to neighbours?	`OLSRegressor` diagnostics (Moran's I on residuals, LM lag/error tests), then spatial lag or spatial error model	Effect of +10% canopy, including the indirect effect on neighbouring cells.
Does that effect differ across the city?	`GWRRegressor`, Geographical Regressor, spatial regimes (north vs south of the Sava), spatial fixed effects (city districts)	Where a tree buys the most cooling.
Which stops are hotter than their surroundings suggest?	`LocalOutlierFactor` with KNN weights	Micro-site problems such as asphalt plazas, glass façades or missing shelter. This becomes a field-inspection list.
How should the city package interventions?	`AgglomerativeClustering` or DBSCAN with regionalization; K-Means for contrast	Contiguous intervention zones with similar heat and vulnerability profiles
Is a stop heat-critical, including planned ones?	SLX, GWR or Geographical classifier	Predicts the thermal class from morphology alone, so it works for future stops and what-if scenarios
Preparing the data	`SpatialImputer` (cloud gaps in surface temperature), spatial and categorical lag transformers, `spatial_train_test_split`, `SpatialPipeline`	Correct features without spatial leakage

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Demo Part 2a: Spatial Dependance Heat has Neighbours

Which ZET stops need shade first?

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Demo Part 2b: Spatial Heterogeneity Where Trees Cool Most

Do trees cool the same everywhere?

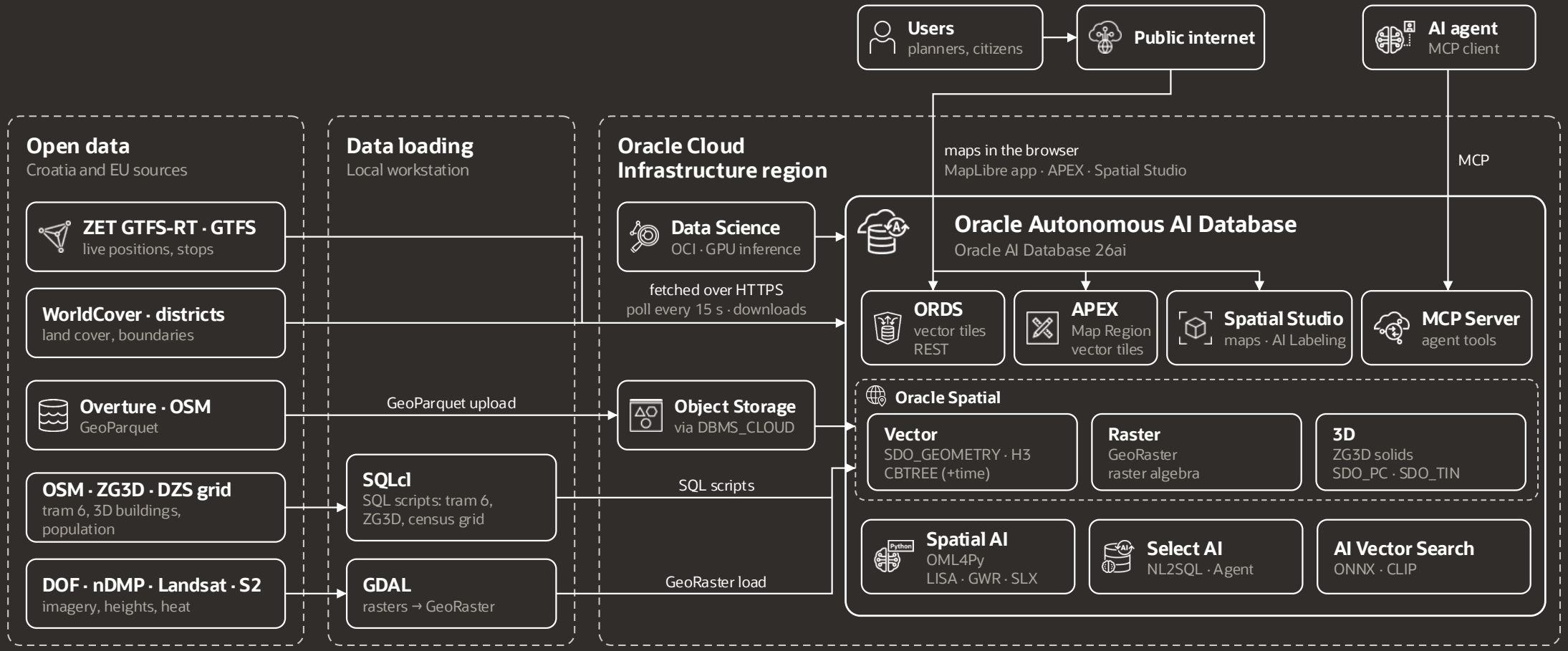
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Demo Part 2c: Spatial Leakage Where the Model Goes Wrong

Can we trust it in a district it has never seen?

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A Solution Architecture
Draft



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Second Direction: Spatial in AI Agents

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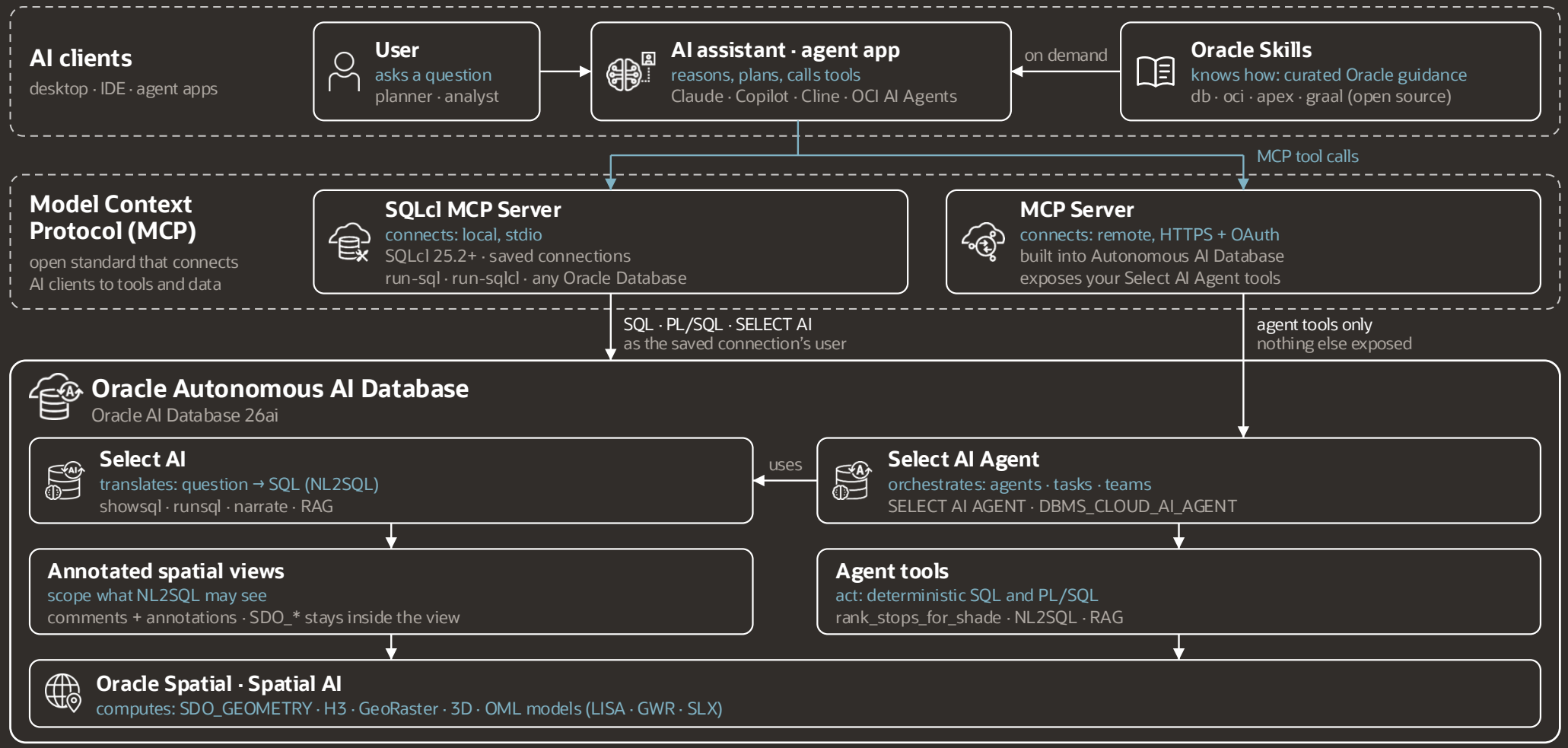
AI, Skills, Agents and MCP on Oracle

From a natural language question to a governed spatial answer

Term	What it does	On Oracle
AI assistant	Reasons, plans and calls tools (the LLM)	Any MCP client: Claude, Copilot, Cline, OCI AI Agents
Skill	Knows *how*: packaged know-how the assistant loads when it needs it	**Oracle Skills** (open source, UPL): `db`, `oci`, `apex`, `graal`
MCP	Connects AI clients to tools and data, as an open standard	**SQLcl MCP Server**: local, any Oracle Database · **MCP Server built into Autonomous AI Database**: remote
NL2SQL	Translates a question into SQL	**Select AI**: `showsql`, `runsql`, `narrate`
Agent	Orchestrates: plans, calls tools, repeats	**Select AI Agent**: agents, tasks, tools, teams
Tool	Acts in a fixed, predictable way	Your SQL and PL/SQL, for example `rank_stops_for_shade(n)`

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AI, Skills, Agents and MCP: How They Fit Together



Every call runs as a database user: privileges, VPD and auditing apply · audit: DBTOOLSMCP_LOG, VCLOUD_AI_SQL

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Demo Part 3:
Ask the Agent: Where Should the Shade Go?

Select AI Agent turns a budget into a plan: One PL/SQL tool ☞ Ranked ZET stops

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SELECT AI AGENT: Components Overview



Tools, tasks, agents and teams are objects in your schema (views USER_AI_AGENT*); every run executes as a database user: privileges, VPD and auditing apply.

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What You (Hopefully) Take Away

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Answers to the Questions

Which ZET stops need shade first?

- Trees go to the green edge, led by Odvojak Reber in Sesvete.
- Shelters go to the busiest central stops: Glavni kolodvor and Trg bana J. Jelačića.

Do trees cool the same everywhere?

- No: 0.47 to 2.13 °C per +10 percentage points of canopy, depending on the neighbourhood

Can we trust it in a district it has never seen?

- Only partly: balanced accuracy drops from 0.85 to 0.72 in Novi Zagreb

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Key Takeaways

- ❑ Keep spatial data and AI in one place
- ❑ Location changes the model
- ❑ Put spatial logic in views and tools, not in prompts
- ❑ Agents stay governed
- ❑ Start small, with open data

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Resources



Oracle Spatial Datasheet



AskTOM: Spatial for DBAs, Analysts and Developers Office Hours



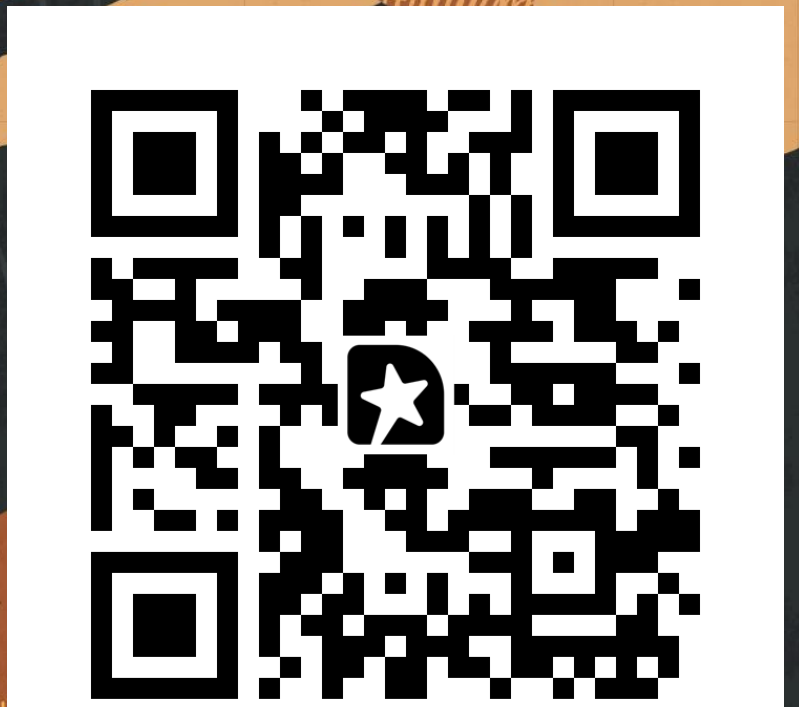
AskTOM: Autonomous AI Database Learning Lounge



Analytics and Data Oracle User Community

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How did you like the journey?



Spatial Meets AI: Intelligent Geospatial Workflows with Oracle

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Q&A

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